Practice workbook · Differentiating twice and proving a relation between y and its derivatives

Differentiating twice (usually with the product rule) and substituting y, dy/dx and d2y/dx2 into a given relation to show it holds or to find constants.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-03

1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

8 Given that $y = e^{3x} \sin 2x$

show that $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$

(8)

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1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 8 continued

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1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 8 continued

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1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 8 marks)

1. Question 8 · part whole · 8 marks · Working and answer

Working and answer

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2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

9 Given that

$$x = e^{-t} \sin 2t$$

show that

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 0$$

(8)

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2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 8 marks)

2. Question 9 · part whole · 8 marks · Working and answer

Working and answer

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3. Question 3 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

3 Given that $y = e^{3x} \sin 2x$

show that $13y + \frac{d^2y}{dx^2} = 6\frac{dy}{dx}$

(8)

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3. Question 3 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 8 marks)

3. Question 3 · part whole · 8 marks · Working and answer

Working and answer

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4. Question 6 · part a.ii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part a.ii only.

6 (a) Given that $y = (4x - 3)e^{2x}$

(i) find $\frac{dy}{dx}$ (3)

(ii) show that $(4x - 3)\frac{dy}{dx} = (8x - 2)y$ (2)

(b) Differentiate $\frac{\sin 5x}{(x-3)^2}$ with respect to x (3)

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4. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

7 $y = e^{2x} \cos 2x$

(a) Show that

$$\frac{dy}{dx} = 2y - 2e^{2x} \sin 2x$$

(4)

(b) Hence show that

$$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$$

(5)

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5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

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5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

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5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 9 marks)

5. Question 7 · part b · 5 marks · Working and answer

Working and answer

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6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

9

$$y = e^{-4t}\cos 2t$$

(a) Show that $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ (3)

Given that $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0$ where M and N are integers

(b) find the value of M and the value of N (5)

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6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

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Question 9 continued

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6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

Question 9 continued

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6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

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Question 9 continued

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(Total for Question 9 is 8 marks)

6. Question 9 · part b · 5 marks · Working and answer

Working and answer

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7. Question 6 · part whole · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

6 $y = e^{4x} \cos 3x$

Show that

$$\frac{d^2y}{dx^2} + Ay = B\frac{dy}{dx}$$ where A and B are integers to be found. **(8)**

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7. Question 6 · part whole · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 8 marks)

7. Question 6 · part whole · 8 marks · Working and answer

Working and answer

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8. Question 5 · part whole · 7 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

5

$$y = e^{2x}(x^2 - 5x)$$

Show that $2e^{2x} = \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y$ (7)

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8. Question 5 · part whole · 7 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

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Question 5 continued

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(Total for Question 5 is 7 marks)

8. Question 5 · part whole · 7 marks · Working and answer

Working and answer

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